MALWANDLA HLONGWANE

ICT Graduate | Software Developer | Android | Database | IT Support & Networking

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PROFESSIONAL SUMMARY

Final-year Diploma in ICT (Applications Development) student at the University of Mpumalanga with hands-on experience across the ICT stack. Built or contributed to seven projects: a live Node.js and MySQL marketplace with 134 automated tests, an enterprise Jakarta EE platform with CI/CD, a secure PHP booking site, an AI-powered Android app, a JavaFX desktop system, a Spring Boot and Flutter app in progress, and a two-campus WAN. Comfortable moving between frontend, backend, mobile, databases and infrastructure. Seeking a graduate programme, internship or junior ICT role.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, PHP, SQL, Dart, HTML5, CSS3, XML
- **Web & Backend:** Node.js, Express.js, Jakarta EE, Spring Boot, REST APIs, MVC and layered architecture, JWT, two-factor authentication
- **Mobile & Desktop:** Android SDK, Android Studio, Room, Retrofit, Material 3, Flutter, JavaFX
- **Databases:** MySQL, PostgreSQL, SQLite; ER modelling, normalisation, indexing, transactions, triggers
- **Testing & DevOps:** Node.js test runner, JUnit 5, Postman, Git, GitHub, GitHub Actions, Docker, Maven, Render
- **Networking & IT Support:** TCP/IP, subnetting, routing and switching, VLANs, wireless LAN, Cisco Packet Tracer, hardware and software troubleshooting, ITIL v3
- **Practices:** Agile/Scrum, SDLC, UML, requirements (SRS), OOP, design patterns, technical documentation, teamwork

PROJECT EXPERIENCE

Taskify | Campus Task & Equipment Marketplace

Full-stack web application | Deployed live
Tech: Node.js, Express, MySQL, JavaScript, HTML, CSS, REST API, JWT, Render

- Built and deployed a full-stack marketplace where students post tasks, hire services, buy items and rent equipment.
- Designed a 26-table MySQL schema (39 foreign keys, 76 indexes) behind a REST API with separate routes, controllers and services.
- Secured the platform with JWT authentication, TOTP two-factor sign-in for admins, bcrypt password hashing, Helmet security headers, rate limiting and server-side input validation.
- Wrote 134 automated tests across 13 test suites (Node.js test runner) covering rentals, sales, disputes and tasks, all passing.

GitHub: github.com/Malwandla21-ict/taskify | **Live site:** taskify-marketplace.co.za

StokVault | Stokvel Contribution & Payout Platform

Enterprise Java | Group project (DICT312) for UMP Centre for Entrepreneurship, Cooperatives & Innovation
Tech: Java 21, Jakarta EE 11 (Faces, REST, EJB, CDI, JPA, JMS, Batch), Payara Server, PostgreSQL, Maven, Docker, GitHub Actions, JUnit 5

- Delivered a layered Jakarta EE application that replaces WhatsApp and notebook record-keeping for stokvels, with role-based access for treasurers, committee members and members.
- Built a SHA-256 hash-chained, tamper-evident audit log, with a PostgreSQL trigger that blocks updates and deletes.
- Set up GitHub Actions CI to build the app, run 49 JUnit 5 unit tests and 48 end-to-end API checks, and build a Docker image.

GitHub: github.com/Malwandla21-ict/StokVault

UMP Events | Campus Event Booking Website
PHP web application | Started as a 2nd-year group project, since rebuilt
Tech: PHP 8.2, MySQL, PDO, MVC, HTML, CSS, JavaScript, Apache (XAMPP)

- Rebuilt a procedural PHP site into a framework-free MVC application with a router, middleware, controllers, services and repositories.
- Prevented overbooking with database transactions, row locks and unique constraints.
- Hardened security against common OWASP risks: prepared statements, CSRF tokens, a strict Content-Security-Policy, secure sessions, login throttling and safe file uploads.

GitHub: github.com/Malwandla21-ict/Ump-Event-Booking-web

HealthWise | AI Symptom Checker Android App
Native Android application
Tech: Java, Android SDK, XML, Material Components 3, Room (SQLite), Retrofit, Gson, AlarmManager, Hugging Face API

- Built a native Android app with a four-step guided symptom check that ranks likely conditions with confidence scores and gives urgency advice.
- Integrated an AI chat assistant through the Hugging Face Inference API using Retrofit, keeping conversation context and suggesting follow-up questions.
- Stored check history, health profile and medication data in Room, with schema migrations that keep user data across upgrades.

GitHub: github.com/Malwandla21-ict/HealthWise

FUEL SA | Fuel Ordering & Station Management System
Java desktop application
Tech: Java 17, JavaFX 21, FXML, CSS, SQLite (JDBC), Maven, BulkSMS REST API

- Built a desktop app where customers register vehicles and request fuel with a live cost estimate, and admins approve or reject requests from a dashboard.
- Integrated the BulkSMS REST API to send request status updates by SMS.

GitHub: github.com/Malwandla21-ict/FuelSA

STAMS | Student Accommodation Management System
Mobile app + REST API | In progress
Tech: Java 21, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, PostgreSQL, Flutter, Dart, Postman

- Building a Flutter mobile app and Spring Boot REST API for landlords to manage properties, rooms and student residents.
- Designing a maintenance workflow where students report issues with photos, track them from Reported to Resolved, and nudge landlords when an issue stalls.

GitHub: github.com/Malwandla21-ict/STAMS-Student-Accomadation-Management-System-

Two-Campus WAN Design & Configuration
Networking project
Tools: Cisco Packet Tracer, TCP/IP, IP addressing & subnetting, routing, switching, wireless LAN

- Designed and configured a wide area network linking two campuses, each with 5 departments of 5 computers.
- Segmented wireless access with separate SSIDs for student, staff and guest networks at both campuses.
- Configured inter-campus routing and verified end-to-end connectivity with ping tests between campuses.

EDUCATION

Diploma in Information and Communication Technology (Applications Development)
University of Mpumalanga | Expected completion: 2026 | Academic average: 75–78%
Relevant coursework:

- **Application Development:** Android (Java, Room, Firebase), Jakarta EE (Servlets, JSF, REST), Web Development (PHP, MySQL, JavaScript)
- **Information Systems III:** ER modelling, normalisation, advanced SQL, transactions, distributed databases, Big Data, NoSQL, BI
- **Networking & Support:** CompTIA A+ and Network+ syllabi, ITIL v3 Foundation
- **Software Engineering:** SDLC, requirements engineering, UML, Agile/Scrum, software testing